EMERGING TRENDS IN SOFTWARE TECHNOLOGIES

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OUTLINE

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EXECUTIVE SUMMARY

- Analysed 65,437 developer survey responses covering languages, databases, and job satisfaction and 110 other attributes.
- This comprehensive report explores the dynamic landscape of the IT industry, providing valuable insights into the most sought-after programming languages, databases, platforms, and web frameworks
- Notably, HTML/CSS, Python, JavaScript, and SQL emerge as the top programming languages in demand, with MySQL, Microsoft SQL Server, PostgreSQL, MongoDB, Redis, and SQLite leading the database landscape.
- Additionally, Google Cloud Platform, AWS, IBM Cloud, Windows, and Microsoft Azure are identified as the top platforms sought after in the industry. React.js, Angular, JQuery, ASP.NET, and Express stand out as the most utilized Web Frameworks.

INTRODUCTION

This comprehensive report delves into the evolving landscape of the IT industry, offering valuable insights into emerging skills and technologies.

The following inquiries were investigated using the data:

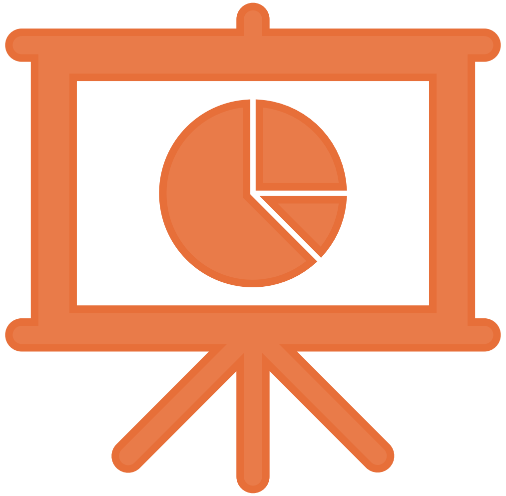
1) Which Programming languages are most in demand today and in the future?

2) Which database technologies are currently most sought currently andin future?

3) What popular IDEs or WEB frames are there?

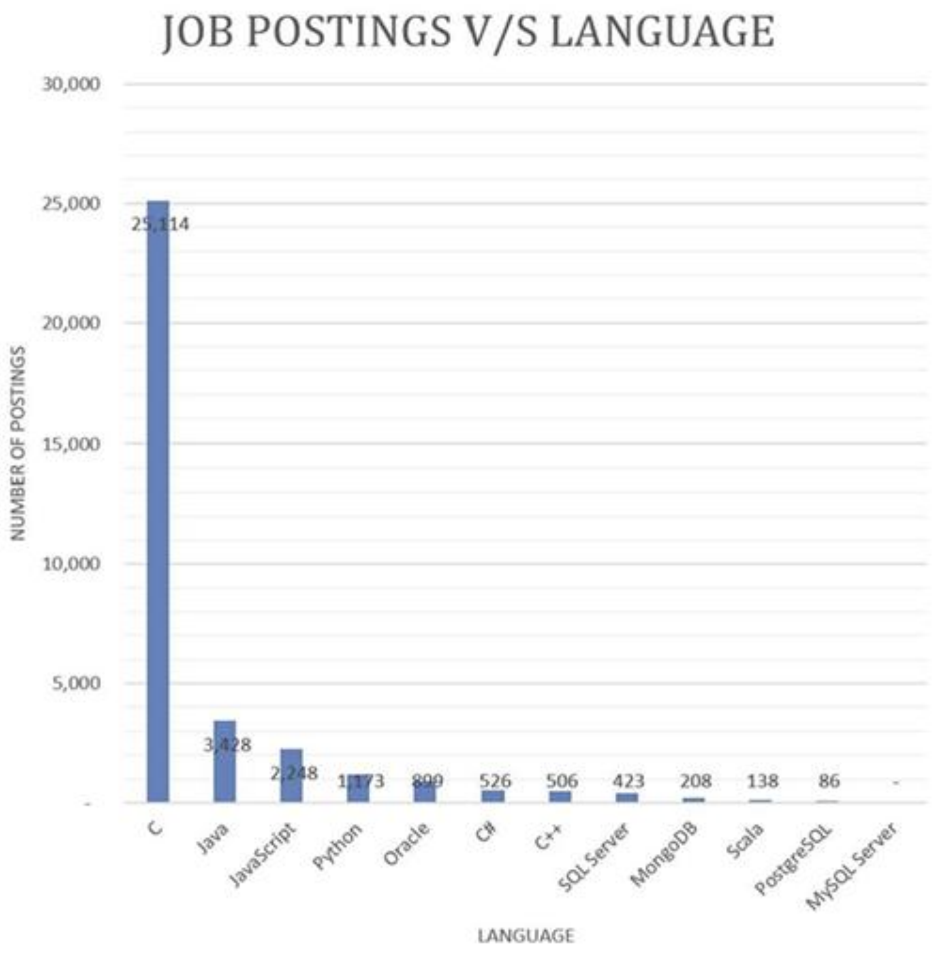
METHODOLOGY

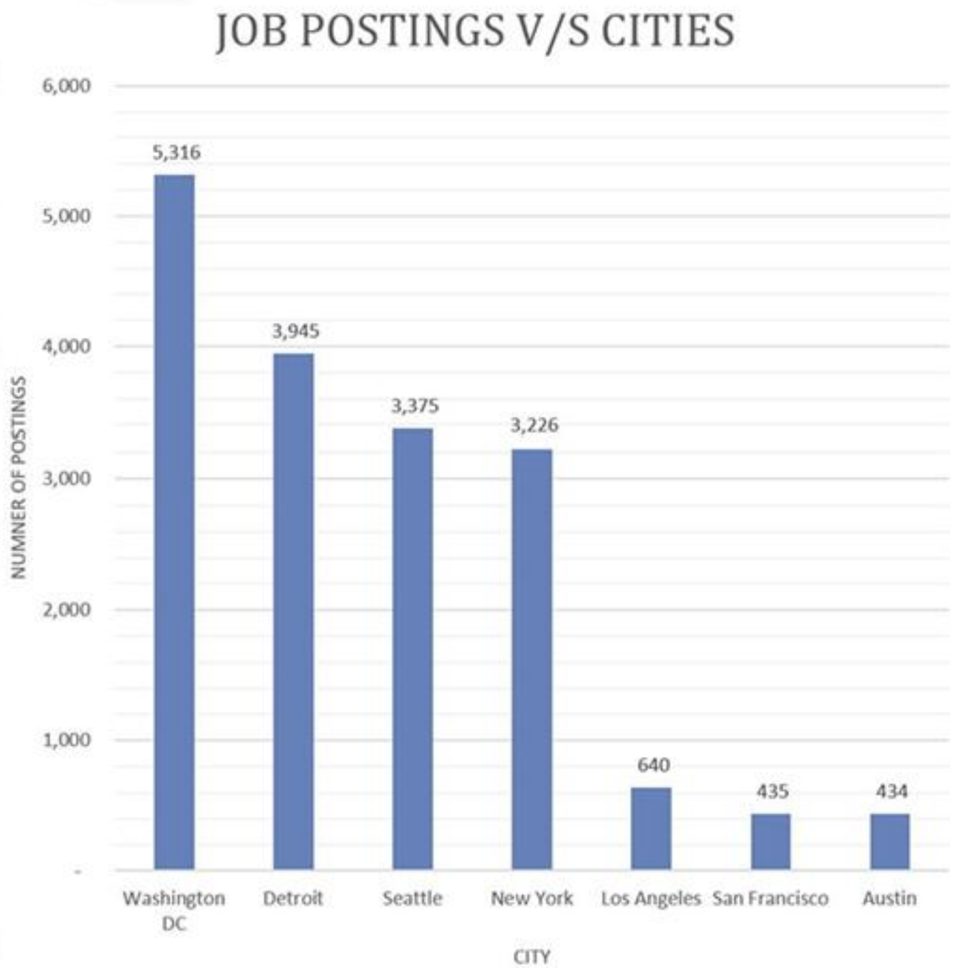
- **Collect survey data**
  - Web scraping
  - APIs
  - Request library
- **Data wrangling:** Cleaned the data for null values, normalised data for future analysis.
- **Exploratory Data Analysis (EDA)**
  - Distribution
  - Outlier Handling
  - Correlation
- **Visualization**
  - Distribution of data
  - Relationship
  - Comparison
  - Composition
- **Dashboards**



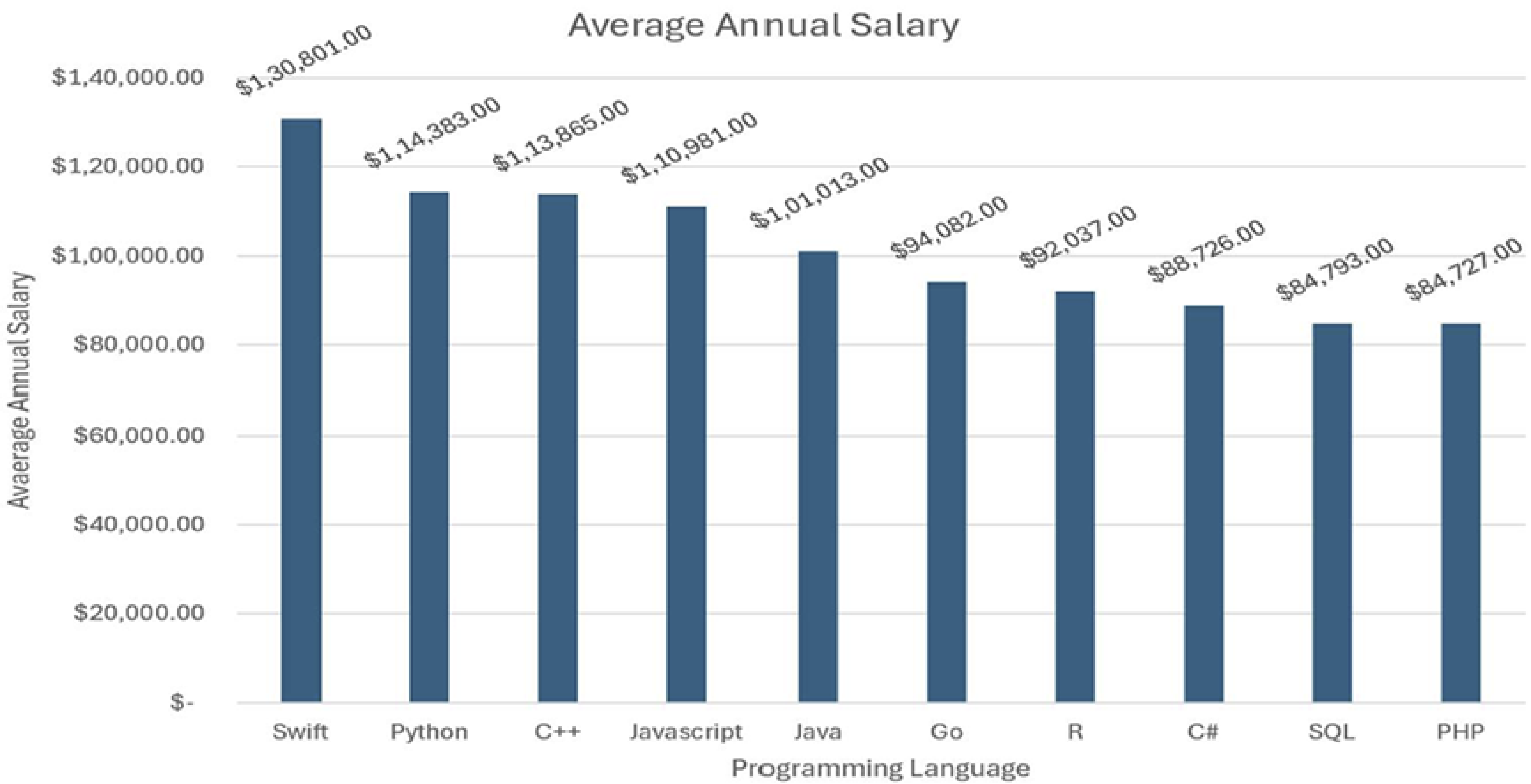
RESULTS →

JOB POSTINGS





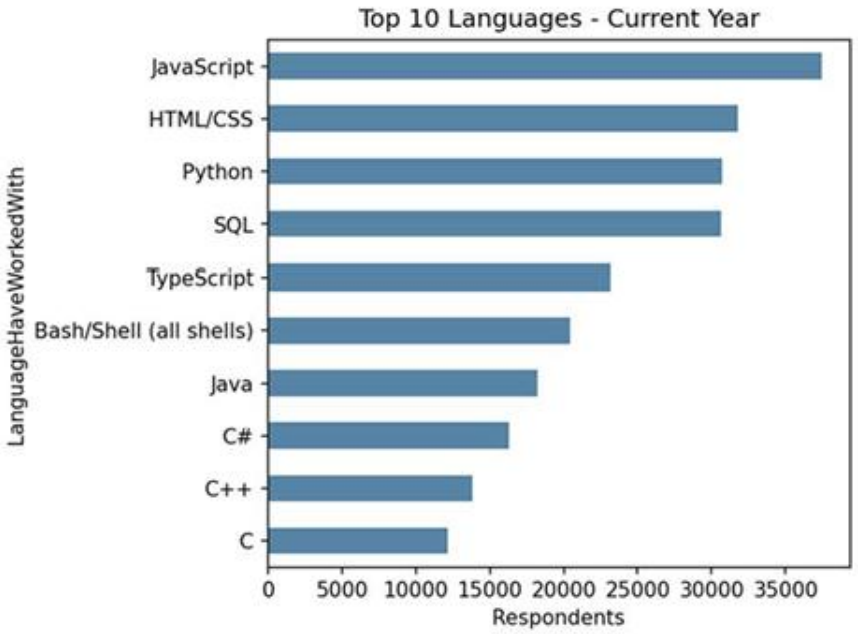
POPULAR LANGUAGES



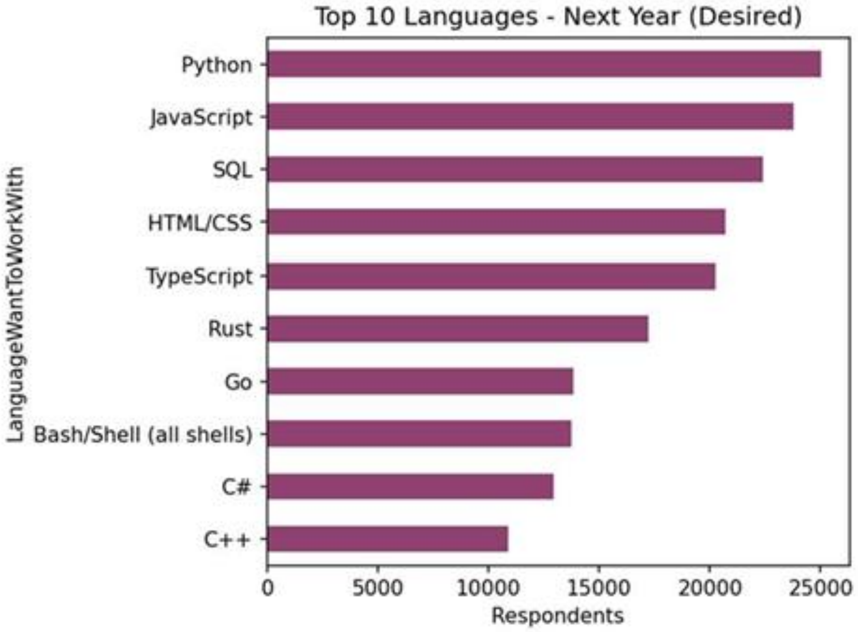
PROGRAMMING LANGUAGE TRENDS

JavaScript turns out to be the most worked langurage for current year, whereas Python emerges as the top upcoming language; Rust and Go show the largest year-over-year growth in interest, signaling a shift toward performance-focused languages.

Current Year



Next Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- JavaScript, HTML/CSS, and Python remain the most widely used languages
- Rust jumped from outside the top 10 currently-used list to #6 in desired languages
- Python holds the top spot for future interest, overtaking JavaScript
- Languages like C and Java find less likings for the future. Respondents are losing interest in Bash and Shell scripting

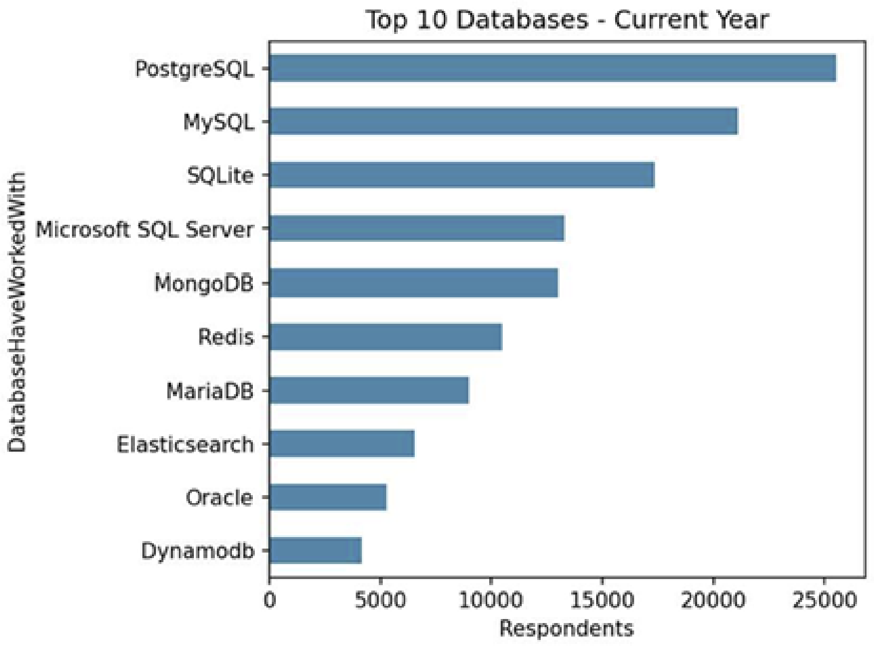
Implications

- Since the web is such a prominent aspect of the tech world it makes sense that web languages are at the top
- Training programs should prioritize Python given its dominance in both current use and future demand
- JavaScript is being continually upgraded and made easier to use, a large complaint is the lack of typing.
- Organizations should monitor Rush adoption as it signals emerging systems-level hiring needs
- Legacy language skills (C, C++) remain relevant but show declining relative interest

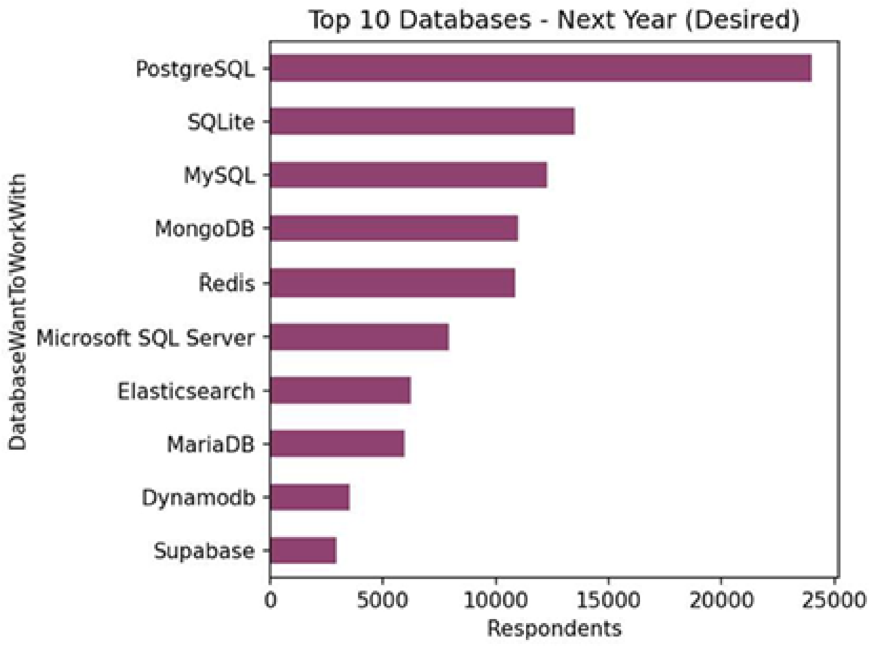
DATABASE TRENDS

PostgreSQL leads in both current use and future demand, while MongoDB and Redis show consistent NoSQL interest, and Supabase emerges as a new entrant in the desired list.

Current Year



Next Year



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- PostgreSQL is the most-used and most-desired database, showing strong sustained demand
- Relational Databases & MySQL usage is high currently but drops in desired-use ranking, suggesting a shift away from it
- NoSQL (Non-relational) databases are increasing in interest such as MongoDB and DynamoDB
- Data stores are increasing in demand such as Redis
- Supabase newly appears in the desired top 10; driving out oracle, signaling growing interest in modern backend-as-a-service tools

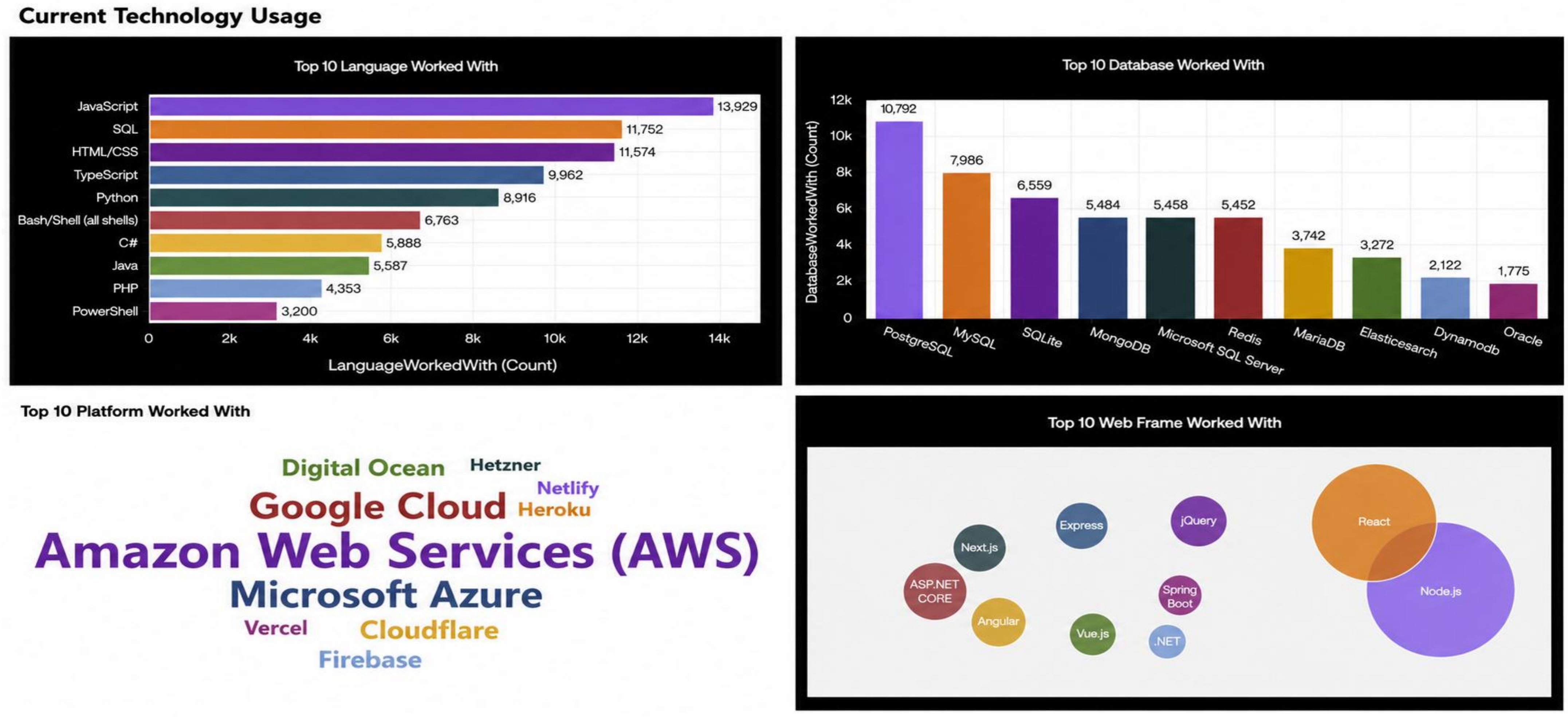
Implications

- PostgreSQL has more features and makes working with SQL easier, which shows a similar trend as with programming languages that simplicity and ease-of-use are important
- With more complex data the freedom that NoSQL provides is very alluring
- Organizations relying heavily on MySQL should plan migration or upskilling paths. Watch Supabase adoption as an early indicator of BaaS platform shifts
- PaaS tools are gaining in popularity, again to simplify the developers life As data becomes more prevalent and valuable storing it becomes more important, thus explaining the increase in data stores
- Organization relying on Oracle DB should make a move as early as possible, as most job-seekers seem to have less inetrest in Oracle DB.

DASHBOARD

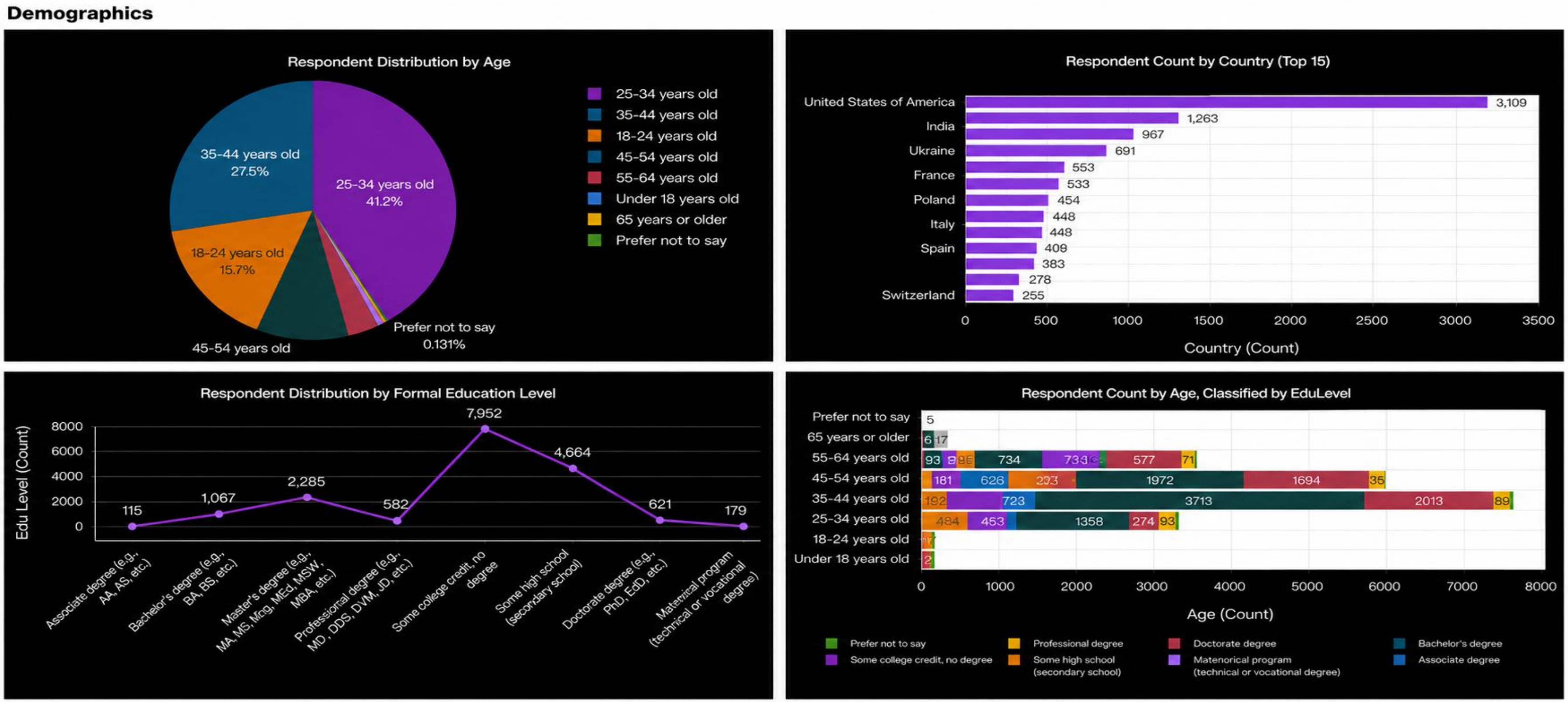
DASHBOARDS →

DASHBOARD TAB 1: Current Technology Usage



DASHBOARD TAB 2: Future Technology Trends

DASHBOARD TAB 3: Demographics



DISCUSSION

- Programming Languages: Strong dominance of Python and JavaScript across demographics.
- Databases: PostgreSQL remains the top choice, with increasing interest in cloud-native databases.
- Demographics: Younger developers favour emerging technologies, while experienced developers prefer stability.

OVERALL FINDINGS & IMPLICATIONS

Findings

- Web development languages emerge as the predominant and sought-after tools within the IT sector presently, highlighting their critical role in driving technological advancements.
- The IT landscape primarily comprises young professionals below the age of 40, suggesting a youthful workforce driving innovation and progress within the industry.
- The desire to acquire skills in Postgre SQL and React JS among most respondents underscores the importance of staying abreast of emerging technologies to remain competitive in the field

Implications

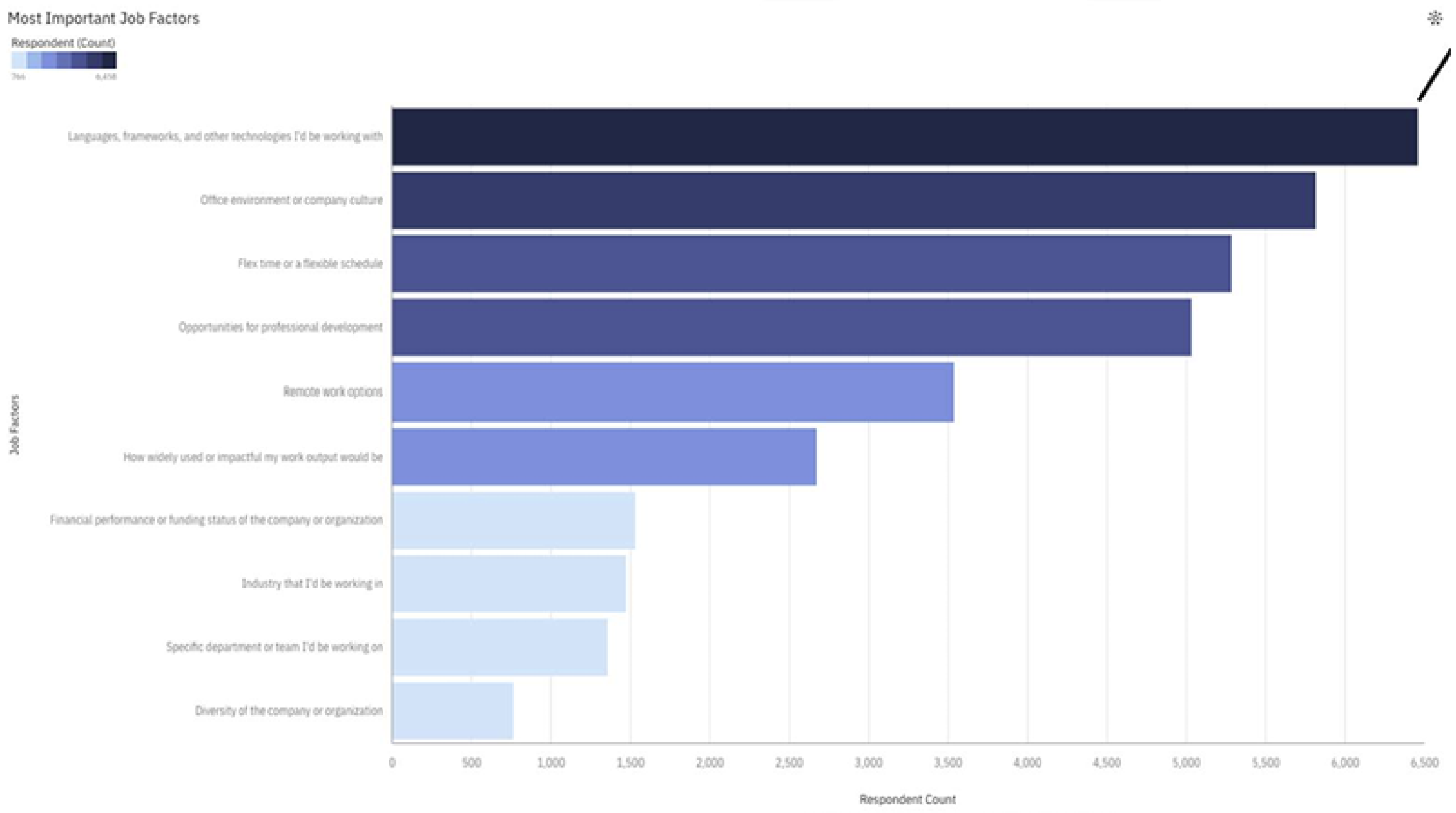
- The respondents being more no the young side could explain the tendency to want to adopt newer languages and technologies
- The continued importance of web development as a lucrative skill underscores the significance of investing in and advancing expertise in this domain.
- Outside of well developed countries, emerging developing countries like Brazil and India are growing in tech training and educations; Less developed countries need more access to tech trainings and educations

CONCLUSION

Developers are looking for 3 main factors when choosing languages and technologies:

1. Languages that are syntactically simple
2. Tools that require less overhead and are more declarative rather than imperative.
3. Tools that are less errorprone which can be accomplished by Typing

APPENDIX



Most Important Job Factor is Coding Technologies Respondents work with!

This is a **Top Factor**.